

- Both the name and place of birth are updated together.
- Place of birth sub-entity is not growing.

One-to-one with References relationship: In this strategy, one data entity references another data entity, where both the entities have a one-to-one relationship with each other.

An example of a one-to-one Referenced relationship, between the user and the details of his place of birth, is illustrated here:

```
{
  "_id": "Ritesh",
  "name": "Ritesh Modi",
  "placeofBirth": "RiteshPOB"
}

{
  "PlaceofBirth": {
    "_id": "RiteshPOB",
    "street": "123 xyz Street",
    "city": "xyzcity",
    "state": "xyzState",
    "zip": "12345"
  }
}
```

One-to-one Referenced relationships should be used when:

- Both the name and place of birth are not retrieved together.
- Both the name and place of birth are updated using different operations.
- Place of birth sub-entity is not growing.

One-to-many with Embedded relationship: In this strategy, a multiple data entity is embedded into another data entity, where they have a one-to-many relationship with each other.

An example of a one-to-many Embedded relationship, between an author and the books he has authored, is illustrated here:

```
{
  "_id": "Ritesh",
  "name": "Ritesh Modi",
  "booksAuthored": [
    {
      "name": "Windows server 2016",
      "publisher": "Self Publishing",
      "year": "2016",
      "price": "30"
    },
    {
      "name": "Ubuntu Linux",
      "publisher": "Self Publishing",

```

```
      "year": "2017",
      "price": "40"
    }
  ]
}
```

One-to-many Embedded relationships should be used when:

- Both the author and the books published are retrieved together frequently.
- Both the author and the books published are updated together.
- Place of birth sub-entity is not growing out of bounds, i.e., there is one-to-few relationship between entities.

One-to-many with References relationship: In this strategy, collections are referenced where they have a one-to-many relationship with each other.

An example of a one-to-many Referenced relationship between authors and books published is illustrated here:

```
{
  "_id": "Ritesh",
  "name": "Ritesh Modi",
}

{
  "_id": "bookid",
  "authorid": "Ritesh",
  "books": [
    {
      "name": "Windows server 2016",
      "publisher": "Self Publishing",
      "year": "2016",
      "price": "30"
    },
    {
      "name": "Ubuntu Linux",
      "publisher": "Self Publishing",
      "year": "2017",
      "price": "40"
    }
  ]
}
```

One-to-many Referenced relationships should be used when:

- Both the author and books published are not retrieved together.
- Both the author and books published are updated at different times in different operations.
- Books authored can grow out of bounds.

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